

Low-Cost EEG and EOG-Based Human–Computer Interaction Using Machine Learning

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Abstract

Electrooculography provides a practical way to use eye activity as an input signal for computer interaction. This paper describes a low-cost EOG-based prototype developed to recognize four ocular states: neutral gaze, upward eye movement, downward eye movement, and blinking. The hardware consists of a BioAmp EXG Pill connected to an ESP32 microcontroller. Vertical EOG is recorded using electrodes positioned above and below the right eye, while the ESP32 samples the signal at 500 Hz, performs initial filtering, and transmits the processed values to a computer.

The available dataset contains 64,000 samples collected across 160 trials, with 40 trials for each class. Every trial lasts 0.8 s and contains 400 samples. A set of 28 handcrafted features was extracted from each trial, covering statistical properties, signal trend, peak behavior, spectral content, and Hjorth mobility. Random Forest, Gradient Boosting, and radial-basis-function Support Vector Machine classifiers were evaluated.

The original implementation used heavily overlapping windows and reported 89.27% five-fold cross-validation accuracy and 91.14% hold-out accuracy with Gradient Boosting. Since neighboring windows shared 87.5% of their samples, this evaluation could overestimate generalization. To obtain a stricter estimate from the same data, the 160 acquisition trials were reconstructed and evaluated directly as independent examples. Under five-fold stratified trial-level cross-validation, Random Forest achieved the highest mean accuracy of **78.75% ± 10.90%**, with a mean macro F1-score of **78.68% ± 10.80%**. Blink and neutral trials were recognized more reliably than the two directional classes, while most directional errors occurred between upward and downward movements.

The classifier was also integrated into a real-time control program. Upward and downward predictions move the cursor vertically, while a double blink is mapped to a mouse click. Prediction confidence, short-term voting, and an action cooldown are used to reduce unstable commands. The results show that a relatively inexpensive EOG setup can support a functioning machine-learning-based interaction prototype. At the same time, the current dataset does not contain participant identifiers, so subject-independent performance cannot be assessed.

Keywords: Electrooculography, EOG, human–computer interaction, blink detection, eye movement classification, machine learning, ESP32, biosignal processing

1. Introduction

Most computer interfaces assume that a user can interact physically with a mouse, keyboard, touchscreen, or controller. That assumption does not always hold. Hands-free interaction can be useful in assistive systems, constrained working environments, and experimental human-machine interfaces. One way to create such an interface is to use physiological signals as an additional input channel.

Brain-computer interface research has established a general processing chain in which physiological activity is acquired, processed, classified, and converted into commands for an external system [Wolpaw et al., 2002]. In this context, electroencephalography and electrooculography should be treated differently. EEG measures electrical activity associated with the brain, while EOG reflects changes in the electrical field around the eye caused by ocular movement. The project described here used both ideas at prototype level, but the EOG component is the part for which a complete dataset, training pipeline, trained model, and evaluation report are available. The quantitative study therefore focuses on EOG.

The electrical potential between the front and back of the eye changes relative to electrodes placed around the eye as gaze direction changes. This makes EOG suitable for detecting intentional eye movements and blinking [Chang, 2019]. Previous studies have used EOG for cursor control, virtual keyboards, mobility systems, communication aids, and other hands-free interfaces [Chang, 2019; Belkhiria et al., 2022].

EOG also comes with practical difficulties. The recorded signal can drift over time, change with electrode placement, pick up muscle and electrical interference, and vary between sessions and users. Blinks may be intentional or involuntary, and small directional movements can be difficult to separate reliably. For these reasons, a useful EOG system requires more than simple signal acquisition. The processing and evaluation strategy matters as much as the hardware.

Several previous systems have used conventional machine-learning methods for this purpose. Ang et al. [2015] demonstrated cursor control using a wearable single-channel EOG interface based on blink patterns. Martínez-Cerveró et al. [2020] presented an open EOG platform for classifying directional eye movements. Hernández Pérez et al. [2023] evaluated wavelet-based EOG features with several supervised classifiers.

The present work explores a smaller command vocabulary using inexpensive hardware. The system recognizes Neutral, Up, Down, and Blink states from vertical EOG recordings and maps those states to a basic computer-control interface.

The main objectives of the study are:

1. to implement vertical EOG acquisition using a BioAmp EXG Pill and ESP32;
2. to collect a balanced four-class EOG dataset;
3. to represent each recorded trial using 28 engineered signal features;
4. to compare Random Forest, Gradient Boosting, and RBF-SVM classifiers;
5. to evaluate performance at trial level rather than relying only on overlapping windows; and
6. to integrate the trained model into a real-time cursor-control prototype.

The work does not introduce a new classifier or a new EOG sensing principle. Its value lies in the complete low-cost pipeline and in the evaluation of how well that pipeline performs on the collected data.

2. Related Work

2.1 EOG as an Input Signal

EOG measures changes in the electrical potential around the eye as the eye moves. A common engineering approximation treats the eye as an electrical dipole, with the corneal side relatively positive compared with the retinal side [Chang, 2019]. When the direction of gaze changes, the voltage measured by nearby electrodes also changes.

Electrode position determines which movements can be measured most directly. Electrodes placed to the left and right of the eyes are commonly used for horizontal movement. Electrodes placed above and below an eye are suitable for vertical movement and also capture strong responses during blinking.

Chang [2019] reviewed the use of EOG in human-computer interaction, including cursor, keyboard, communication, and mobility applications. Belkhiria et al. [2022] reviewed EOG-HCI work published over two decades and described acquisition, preprocessing, feature extraction, classification, and command translation as recurring components of these systems.

The architecture used in this project follows the same general idea, although the implementation is intentionally small and low-cost.

2.2 EOG for Cursor Control

Cursor control is a natural EOG application because directional movement can be translated into navigation and blinking can be translated into a selection command.

Ang et al. [2015] developed a wearable single-channel EOG interface that used blink patterns for cursor operation. Martínez-Cerveró et al. [2020] investigated directional EOG classification using an open software/hardware platform and supervised learning.

These studies are relevant to the present work because they show that a limited set of ocular gestures can be converted into practical computer commands. However, direct comparison of accuracy values would be misleading because the hardware, classes, participants, features, and validation procedures differ.

2.3 Conventional Classifiers for EOG

The current project compares three established models rather than introducing a new learning algorithm.

Random Forest combines multiple decision trees and aggregates their predictions [Breiman, 2001]. It is useful for nonlinear relationships and mixed feature types.

Gradient Boosting builds decision-tree models sequentially, where later trees focus on errors left by earlier ones [Friedman, 2001].

Support Vector Machines construct separating boundaries in feature space and can represent nonlinear decision surfaces through kernel functions [Cortes and Vapnik, 1995].

Hernández Pérez et al. [2023] also used conventional supervised algorithms for EOG movement classification, showing that engineered features and standard classifiers remain relevant for this type of problem.

2.4 Feature Representation

A single amplitude threshold does not capture every property of an EOG waveform. The current system therefore combines several kinds of information: amplitude statistics, waveform slope, zero crossings, peak structure, temporal changes, spectral components, and Hjorth mobility.

Hjorth mobility was introduced as a time-domain descriptor for electrophysiological signals [Hjorth, 1970]. In this work it is used simply as another measure of waveform behavior, without assigning a physiological interpretation to it.

3. Research Problem

The task is to classify a short vertical-EOG recording into one of four classes.

For a signal segment x , the feature extraction function produces

$$\mathbf{z} = \phi(x) \in \mathbb{R}^{28}.$$

The classifier then predicts

$$\hat{y} = f(\mathbf{z}),$$

where

$$\hat{y} \in \{Neutral, Up, Down, Blink\}$$

The study addresses the following questions.

How accurately can complete EOG trials be classified into the four target classes?

Which of Random Forest, Gradient Boosting, and RBF-SVM performs best under trial-level cross-validation?

RQ3

Do different groups of handcrafted features provide complementary information for classification?

RQ4

Can the resulting model be used inside a real-time control program for basic computer interaction?

The fourth question concerns implementation rather than a formal user study. No end-to-end latency or usability metric is claimed.

4. System Architecture

The implemented signal path is:

Eye movement → Electrodes → BioAmp EXG Pill → ESP32 → Serial transmission → Computer processing → C

4.1 Hardware

The BioAmp EXG Pill provides the analog physiological-signal input. Its output is connected to GPIO 34 of the ESP32.

The electrode arrangement is vertical:

- **IN+** above the right eye;
- **IN-** below the right eye;
- **REF** near the earlobe or mastoid.

This placement supports vertical eye movement and blink detection. The supplied model does not contain LEFT or RIGHT classes, so horizontal cursor movement is not included in the evaluated system.

4.2 Sampling and Serial Communication

The ESP32 samples the signal at

$$f_s = 500 \text{ Hz.}$$

The nominal sampling period is

$$T_s = \frac{1}{500} = 0.002 \text{ s.}$$

The ADC is configured to 12-bit resolution.

Serial communication runs at 115200 baud, and every transmitted record contains:

5. Embedded Signal Processing

5.1 ADC Voltage Conversion

The raw ADC value is mapped from the range 0–4095 to a 3.3 V scale and then centered around the mid-rail value:

$$V[n] = \frac{ADC[n]}{4095} \times 3.3 - 1.65.$$

5.2 High-Pass Stage

A first-order high-pass filter is used to reduce slow drift:

$$y_{HP}[n] = \alpha^{HP} y_{HP}[n-1] + (1 - \alpha^{HP}) x[n].$$

The firmware uses

$$\alpha_{HP} \approx 0.99373,$$

derived from a nominal cutoff of about 0.5 Hz.

5.3 Low-Pass Stage

The next stage is a first-order low-pass filter:

$$y_{LP}[n] = \alpha_{LP} y_{LP}[n-1] + (1 - \alpha_{LP}) x[n],$$

with

$$\alpha_{LP} \approx 0.27363.$$

The nominal cutoff is approximately 30 Hz.

5.4 Baseline Correction

A moving average of 500 samples is maintained:

$$B[n] = \frac{1}{500} \sum_{k=n-499}^n y_{LP}[k].$$

The baseline-corrected signal is

$$x_{out}[n] = y_{LP}[n] - B[n].$$

Because the sampling frequency is 500 Hz, the baseline window represents approximately one second.

6. Data Collection

6.1 Classes

Four labels were used.

ID	Class	Instruction
0	Neutral	Look straight ahead and keep the eyes still
1	Up	Look upward deliberately
2	Down	Look downward deliberately
3	Blink	Perform one natural blink

The acquisition sequence was randomized.

6.2 Dataset Size

The collection script was configured for 40 trials per class.

Therefore,

$$40 \times 4 = 160$$

trials were recorded.

Each trial contains 400 samples:

$$400/500 = 0.8 \text{ s.}$$

The complete dataset therefore contains

$$160 \times 400 = 64,000$$

samples.

Class	Trials	Samples
Neutral	40	16,000
Up	40	16,000

Class	Trials	Samples
Down	40	16,000
Blink	40	16,000
Total	160	64,000

The CSV stores:

- timestamp,
- raw ADC value,
- filtered voltage,
- class ID,
- class name.

6.3 Trial Reconstruction

The CSV does not contain an explicit trial identifier. However, the acquisition code records exactly 400 samples for each trial. The 64,000-row dataset can therefore be divided into 160 consecutive blocks of 400 samples.

Each reconstructed block was checked for label consistency and treated as one trial for the stricter evaluation described later.

Participant identity is not available in the stored files. For this reason, the study does not make claims about generalization to unseen users.

7. Computer-Side Preprocessing and Feature Extraction

The training pipeline applies an additional 0.5–35 Hz band-pass filter followed by a 50 Hz notch filter before feature extraction.

Each 400-sample trial is converted into 28 numerical features.

7.1 Statistical Features

The first group contains:

- mean,
- standard deviation,
- minimum,
- maximum,

- peak-to-peak range,
- RMS,
- skewness,
- kurtosis,
- sum of absolute amplitudes,
- mean absolute derivative.

For example,

$$RMS = \sqrt{\frac{1}{N} \sum_{n=1}^N x_n^2}.$$

Peak-to-peak range is

$$R = \max(x) - \min(x).$$

The mean absolute first difference is

$$D = \frac{1}{N-1} \sum_{n=2}^N |x_n - x_{n-1}|.$$

7.2 Trend and Zero-Crossing Features

A linear slope is fitted across the signal window.

The zero-crossing rate is

$$ZCR = \frac{1}{N} \sum_{n=2}^N I[\text{sign}(x_n) \neq \text{sign}(x_{n-1})].$$

7.3 Peak Features

The extraction code measures:

- number of positive peaks,
- number of negative peaks,
- largest positive peak,
- smallest negative peak.

7.4 Temporal Segment Features

The window is divided into three approximately equal regions.

Their means are denoted

$$\mu_1, \mu_2, \mu_3.$$

A final temporal descriptor is

$$\mu_3 - \mu_1.$$

This gives the classifier some information about how the signal changes over the duration of the trial.

7.5 Frequency-Domain Features

A real-valued FFT is calculated for the window.

Power descriptors are obtained over:

- 0–2 Hz,
- 2–8 Hz,
- 8–15 Hz,
- 15–30 Hz.

Relative values for the first two ranges and the dominant non-zero frequency are also included.

These ranges are used only as numerical features for EOG classification. They are not interpreted as direct measures of cognitive EEG activity.

7.6 Hjorth Mobility

The final feature is Hjorth mobility:

$$M = \sqrt{\frac{\text{Var}(\Delta)x}{\text{Var}(x) + \epsilon}}.$$

8. Machine-Learning Models

8.1 Random Forest

The Random Forest configuration uses:

- 300 trees,
- maximum depth 15,
- minimum leaf size 2,
- balanced class weights,
- random state 42.

8.2 Gradient Boosting

The Gradient Boosting model uses:

- 200 estimators,
- learning rate 0.08,
- maximum depth 5,
- subsample 0.8,
- random state 42.

8.3 RBF-SVM

The Support Vector Machine uses:

- RBF kernel,
- $C=10$,
- scaled gamma,
- balanced class weighting,
- probability estimation.

8.4 Scaling

The models are placed inside pipelines that standardize the features:

$$z_j = \frac{x_j - \mu_j}{\sigma_j}.$$

9. Evaluation Methodology

Two evaluation strategies are reported separately.

9.1 Original Overlapping-Window Evaluation

The original training pipeline used 400-sample windows with a 50-sample step.

The overlap was therefore

$$400 - 50 = 350$$

samples, or

$$\frac{350}{400} \times 100 = 87.5\%.$$

Using this pipeline, Gradient Boosting achieved:

- five-fold cross-validation accuracy: **89.27%**;
- hold-out accuracy: **91.14%**.

The hold-out class metrics were:

Class	Precision	Recall	F1	Support
Neutral	1.00	0.86	0.93	22
Up	0.81	0.94	0.87	18
Down	0.89	0.85	0.87	20
Blink	0.95	1.00	0.97	19

Its confusion matrix was

[19 1 1 10 17 1 00 3 17 00 0 0 19].

These numbers are retained because they are part of the original implementation, but they are not used as the strongest evidence of generalization.

9.2 Trial-Level Evaluation

For the stricter analysis, the 160 reconstructed acquisition trials were treated as independent examples.

Five-fold stratified cross-validation was performed using complete trials.

The same feature set and model configurations were retained.

The main metrics were accuracy and macro F1.

$$Accuracy = \frac{\text{correct predictions}}{\text{total predictions}}.$$

For each class,

$$Precision = \frac{TP}{TP + FP},$$

$$Recall = \frac{TP}{TP + FN},$$

and

$$F1 = \frac{2PR}{P + R}.$$

Macro F1 is the arithmetic mean of the four class-level F1 values.

10. Results

10.1 Model Comparison

The trial-level results were lower than the original overlapping-window results.

Classifier	Accuracy, mean $\pm$ SD	Macro F1, mean $\pm$ SD
Random Forest	78.75% $\pm$ 10.90%	78.68% $\pm$ 10.80%
Gradient Boosting	73.75% $\pm$ 3.19%	73.82% $\pm$ 3.76%
RBF-SVM	71.25% $\pm$ 6.67%	70.70% $\pm$ 6.91%

Random Forest achieved the highest mean trial-level accuracy.

This ranking is different from the original overlapping-window evaluation, where Gradient Boosting performed best.

The result shows that the choice of evaluation unit affects both the absolute accuracy and the apparent ranking of the models.

10.2 Class-Level Random Forest Results

Pooling predictions from the five folds gave:

- overall accuracy: **0.7875**;
- macro precision: **0.7848**;
- macro recall: **0.7875**;
- macro F1: **0.7845**.

Class	Precision	Recall	F1	Support
Neutral	0.80	0.90	0.85	40
Up	0.78	0.73	0.75	40
Down	0.72	0.65	0.68	40
Blink	0.83	0.88	0.85	40

Neutral and Blink were the strongest classes in this evaluation.

Down had the lowest recall.

10.3 Confusion Matrix

The pooled confusion matrix was

[36 1 1 2 2 29 7 2 6 5 26 3 1 2 2 35].

Actual \ Predicted	Neutral	Up	Down	Neutral	Blink
Neutral	36	1	1	2	2
Up	2	29	7	2	2
Down	6	5	26	1	2
Neutral	36	2	2	3	35

The main directional error was confusion between Up and Down.

11. Feature-Group Analysis

A small feature-group study was also performed using Random Forest and the same trial-level folds.

The tested groups were:

- statistical features;
- trend, peak, and segment features;
- frequency-domain features;
- time-domain features plus Hjorth mobility;
- all 28 features.

Feature group	Accuracy, mean $\pm$ SD	Macro F1
Statistical	76.25% $\pm$ 11.46%	0.760
Trend + peak + segment	68.13% $\pm$ 9.56%	0.674
Frequency-domain	75.63% $\pm$ 4.59%	0.753
Time-domain + Hjorth	75.00% $\pm$ 7.65%	0.748
All 28 features	78.75% $\pm$ 10.90%	0.785

The complete feature set produced the highest mean accuracy.

The difference was not large enough to claim that every feature is necessary. Statistical and spectral features alone already retained much of the classification performance.

12. Real-Time Control

The trained model is loaded by a Python program that continuously receives EOG samples over serial communication.

12.1 Rolling Window

The program stores the latest 400 samples and updates its prediction after every 50 new samples.

At 500 Hz, this gives a nominal update interval of

$$50/500 = 0.1 \text{ s.}$$

This is the model-update interval, not a measured end-to-end response time.

A prediction is allowed to affect the interface only when its class probability is at least

$$0.55.$$

Lower-confidence outputs are ignored.

12.3 Short-Term Voting

The four most recent predicted labels are stored, and the most frequent label is used as the stable prediction.

$$\hat{y}_{\text{stable}} = \text{mode}(\hat{y}_{t-3}, \hat{y}_{t-2}, \hat{y}_{t-1}, \hat{y}_t).$$

12.4 Command Mapping

The class-to-command mapping is:

Class	Action
Neutral	No movement
Up	Move cursor upward
Down	Move cursor downward
Blink	Update blink state

Directional movement uses a default cursor step of 30 pixels.

A 0.25 s cooldown is used to prevent repeated actions from being triggered too quickly.

12.5 Double Blink

A blink is stored temporarily.

If a second blink is detected within 700 ms of the first, the program sends a mouse click.

Thus, the machine-learning model identifies individual blink events, while the interface logic determines whether those events form a double blink.

13. Broader Prototype Context

The original engineering project also included demonstrations beyond the EOG classifier.

These included:

- blink-based interaction with a browser game;
- focus-based racing-game control;
- focus-based relay switching for an electrical light;
- EOG cursor control.

The available focus-controller code reads a numerical focus value and compares it with a threshold. The files available for this paper do not contain the complete method used to calculate that value from raw EEG.

For that reason, the EEG-based demonstrations are not included in the quantitative evaluation.

14. Discussion

The most important result is not simply the highest accuracy value. The comparison between the two evaluation strategies gives a more useful picture of the system.

The original pipeline reported 91.14% hold-out accuracy, but those windows overlapped by 87.5%. Two neighboring windows could therefore contain mostly the same signal.

For example,

$$W_1 = x[1 : 400]$$

and

$$W_2 = x[51 : 450]$$

share 350 samples.

If such windows are separated across training and testing, the resulting test set is not fully independent of the training data.

When complete acquisition trials were treated as the evaluation units, the best mean accuracy fell to 78.75%.

That lower result is less visually impressive, but it is more informative. It suggests that the original overlapping-window experiment was easier than classifying separate recorded trials.

The change in model ranking is also worth noting. Gradient Boosting was strongest in the original pipeline, whereas Random Forest performed better under the trial-level analysis. This does not establish that Random Forest is generally better for EOG. It simply shows that the preferred model can depend on how the data is divided.

The class results also show that all four classes were not equally difficult. Neutral and Blink were classified more reliably, while Up and Down produced more mutual errors.

This is plausible given that both directional states were recorded using the same vertical electrode pair. However, the dataset does not allow the cause of those errors to be isolated. They may reflect differences in movement timing, electrode contact, signal variability, preprocessing, or limitations of the feature representation.

The feature-group analysis gives a similar caution. The full 28-feature vector performed best, but the advantage over the statistical and frequency-only groups was modest. The result supports using mixed features in this dataset, but it does not establish an optimal feature set.

The real-time control layer is also important because a classification model by itself is not a complete interface. A single false prediction can immediately become an unwanted cursor movement. For this reason, the implementation uses a confidence threshold, a short voting buffer, and an action cooldown.

These mechanisms reduce unstable commands, although they may also increase effective response time. Because end-to-end latency was not measured, the current study does not claim a specific interaction speed.

Overall, the prototype demonstrates that the acquired EOG signal contains enough information to support a small four-class control vocabulary. The remaining question is how well the same approach would work across different people and recording sessions, which cannot be answered from the current dataset.

15. Limitations

15.1 Participant Information

The dataset does not contain participant identifiers.

Although 160 trials can be reconstructed, the number of people who contributed to those recordings cannot be established from the stored files.

Subject-independent performance is therefore unknown.

15.2 Trial IDs Were Not Stored

The original CSV contains class labels but no explicit trial identifier.

The trial-level analysis was possible because the collection protocol recorded exactly 400 samples per trial and the dataset could be divided into consistent 400-sample blocks.

Future data collection should store trial IDs directly.

15.3 Dataset Size

The study contains 40 trials for each of four classes.

This is adequate for a preliminary comparison but still small for making broad claims about generalization.

The Random Forest cross-validation accuracy also varied considerably across folds:

$$78.75\% \pm 10.90\%.$$

This variation suggests that the result is sensitive to the particular trials used for training and testing.

15.4 No Subject-Independent Evaluation

Leave-one-subject-out evaluation cannot be performed because participant identity was not retained.

It is therefore unknown whether a model trained on the current recordings would work for a new user without retraining or calibration.

15.5 Limited Directional Vocabulary

The trained classifier contains only:

- Neutral,

- Up,
- Down,
- Blink.

The current paper therefore does not report LEFT or RIGHT classification.

15.6 Difference Between Offline and Real-Time Filtering

The training code uses forward-backward filtering, while the online controller uses causal streaming filtering.

The two approaches do not produce perfectly identical waveforms.

A later version should use the same causal preprocessing during both training and deployment.

15.7 End-to-End Latency Was Not Recorded

The 100 ms prediction-update interval is determined by the 50-sample step size.

It does not include the full delay introduced by signal acquisition, window formation, filtering, feature extraction, classification, voting, cooldown logic, and cursor execution.

No measured end-to-end latency is available.

15.8 No Controlled User Study

The prototype was used as a working interface, but no structured study measured command success, false activations, click success, or task-completion time.

The present results should therefore be interpreted as classifier and implementation results rather than a full usability evaluation.

16. Future Work

16.1 Multi-Participant Recording

A future version of the collection program should store:

- participant ID,
- session ID,
- trial ID,
- class,
- timestamp,

- raw signal,
- processed signal.

This would make the dataset much easier to analyze and reproduce.

16.2 Subject-Independent Evaluation

Once recordings from several participants are available, leave-one-subject-out evaluation would provide a more meaningful measure of generalization.

For each fold, the system could be trained on all but one participant and tested on the excluded participant.

16.3 Consistent Preprocessing

The online causal filtering pipeline should also be used during model training so that both stages operate on the same type of signal.

16.4 Horizontal EOG

A separate horizontal electrode configuration could be recorded to add LEFT and RIGHT classes.

The extended class set could then become:

$$\{Neutral, Up, Down, Left, Right, Blink\}$$

Only data actually recorded from such a configuration should be used to evaluate that extension.

17. Conclusion

A low-cost EOG-based interaction system was developed using a BioAmp EXG Pill, ESP32 microcontroller, conventional signal processing, engineered features, and supervised machine learning.

The available dataset contains 160 balanced trials and 64,000 signal samples across Neutral, Up, Down, and Blink classes.

Each trial was represented by 28 statistical, temporal, peak, spectral, and Hjorth-related features.

The original overlapping-window pipeline reported 89.27% cross-validation accuracy and 91.14% hold-out accuracy with Gradient Boosting. A stricter trial-level analysis of the same recorded data produced a lower but more defensible result. Random Forest achieved

78.75% $\pm$ 10.90%

mean five-fold accuracy and

78.68% $\pm$ 10.80%

mean macro F1.

Neutral gaze and blinking were classified more reliably than the two directional movements, while the largest source of confusion was between Up and Down.

The same EOG classes were connected to a real-time cursor-control program. Up and Down predictions produced vertical movement, and two detected blinks within 700 ms produced a mouse click. Confidence filtering, short-term voting, and an action cooldown were used to reduce unstable commands.

The results support the feasibility of the current prototype, but they should not be treated as evidence of user-independent assistive performance. The next meaningful improvement is a multi-participant dataset with explicit participant and trial identifiers, followed by subject-independent evaluation.

Declarations

DataAvailability

The study uses the EOG dataset generated during development of the prototype. The available dataset contains 64,000 labeled samples across four classes.

A permanent public repository link should be added if the final publication venue requires open data.

Code Availability

The project includes separate implementations for acquisition, training, and real-time cursor control.

A repository or archival link should be added before submission if required by the target venue.

Ethics Statement

The available project records do not contain sufficient information to establish the participant count, consent procedure, or institutional ethics requirements that applied during data collection.

This information must be confirmed by the authors before submission.

Conflict of Interest

[To be completed by the authors.]

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